Supply Chain Logistics Analysis

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DSA3050: Business Intelligence and Data Visualization

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Part A: Data Acquisition and Understanding (10 Marks)

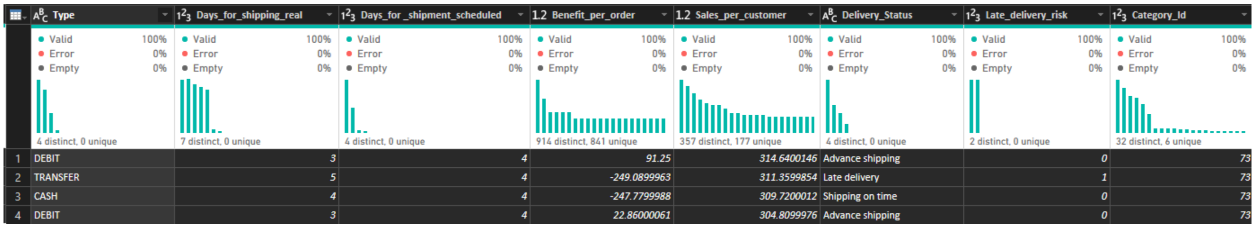
The selected raw dataset has 53 columns and 180,519 rows on the supply chain of a shipping company. This dataset examines the payment methods, shipping destinations, order date and time, shipping date and time, and the like. Moreover, it has both categorical and numerical variables with two dedicated date fields. One pertaining to the order date and another pertaining to the shipping date. This enables 16 dimension tables to be created. This dataset was obtained from Kaggle under a CC0 (Creative Commons 0) license and was originally retrieved from Mendley Data under the same license.

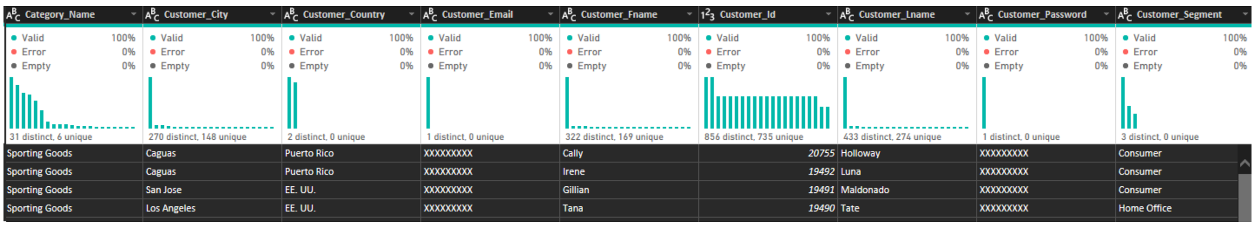
This raw dataset has the key columns of:

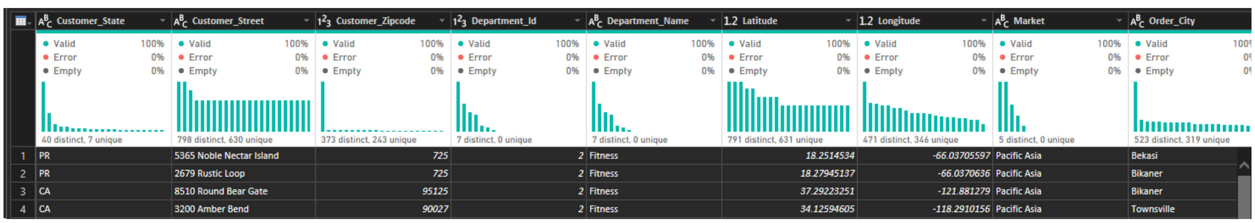
- **Type –** The payment method used. This could inform whether delivery speed could be correlated with the means an individual pays. Although, this is a highly unlikely correlation.
- **Days_for_shipping_real –** The actual time it took to ship the products
- **Days_for_shipping_scheduled –** The time slotted to ship the product.
- **Delivery_Status –** The status of the delivery derived in part from the amount of time it took to ship the product viz a viz how long it was supposed to take.
- **order_date_DateOrders –** The date and time that a product was ordered.
- **shipping_date_DateOrders –** The date and time that a product was dispatched for shipment.
- **Shipping_Mode –** The priority rank of a shipment. There might be a correlation between this priority and whether a shipment is early or late.
- **Order_Profit_Per_Order –** The amount of money made net for the shipment. This value can be negative indicating the the company made a loss. There might be a correlation between how much profit is made and how much priority a product is given in shipment.

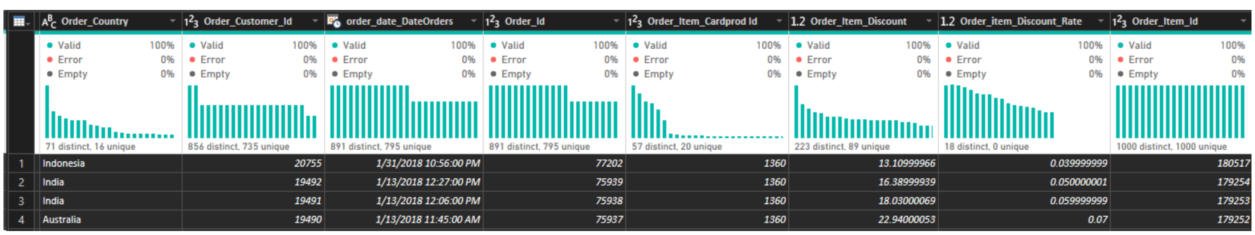
My chosen domain is Supply Chain Logistics Analysis. My business intelligence analysis will evaluate the speed of delivery of orders across different countries, departments, product lines, and market segments in order to uncover the general performance of the supply chain, the exact gaps for the company, and what factors correlate to cause either timely, rapid, or late shipping.

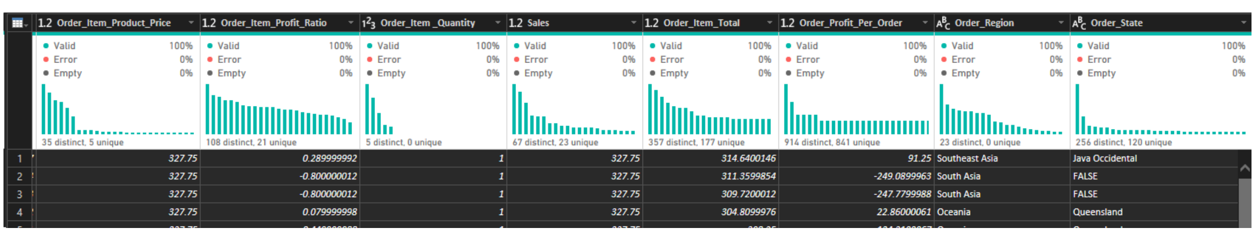
This dataset is suitable for this analysis since it exposes a rich array of data that is invaluable to stakeholders who would want to understand the company's delivery patterns (if it stayed consistent or if it fluctuated) and what company policies to institute in order to ensure that they greatly minimize the late delivery patterns.

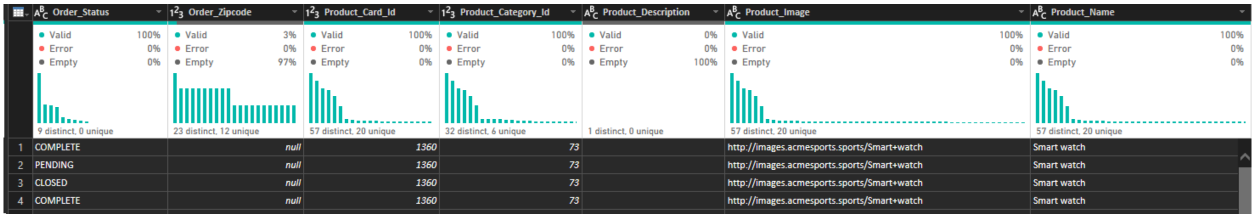


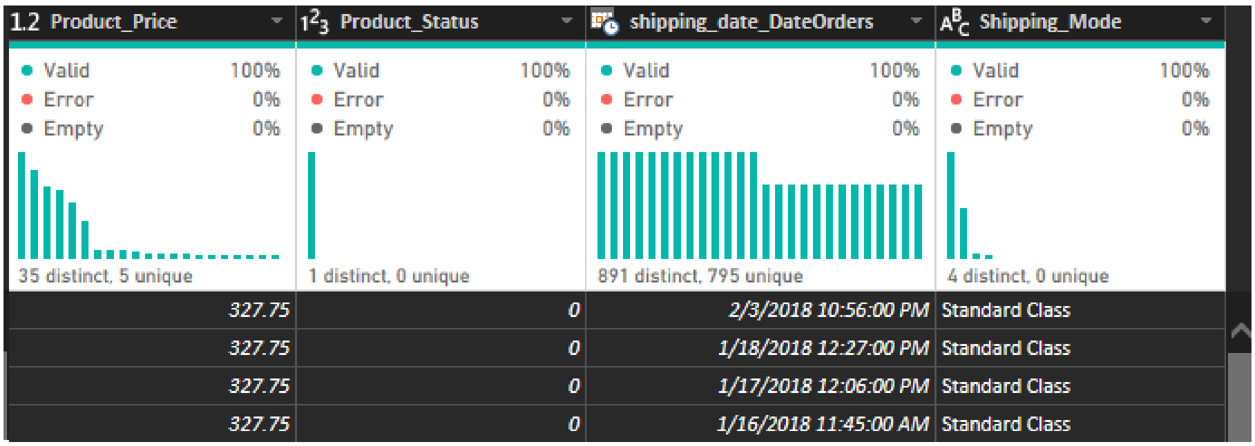






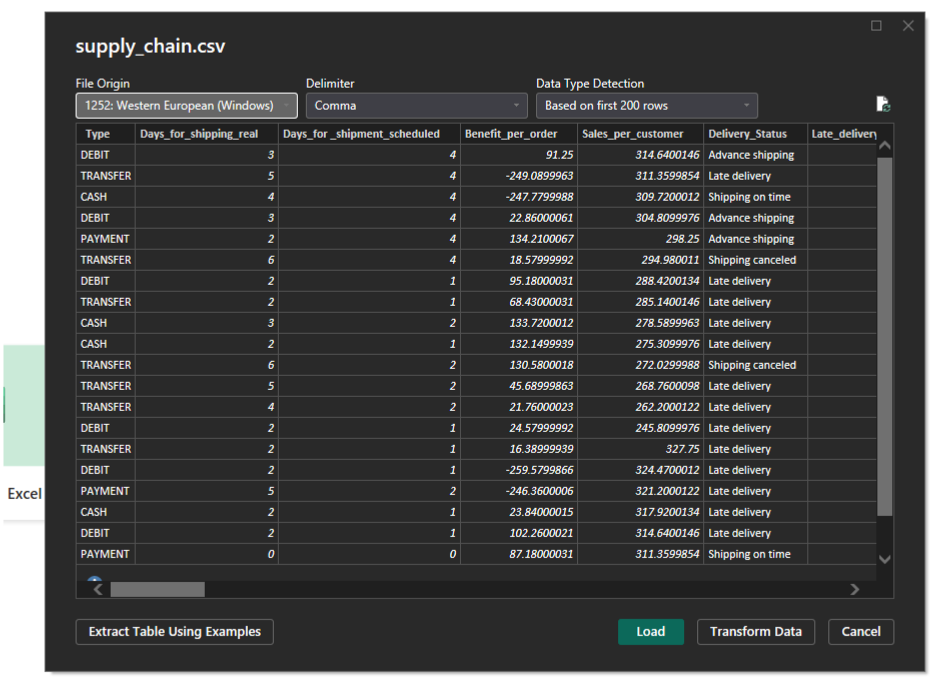






Part B: Data Cleaning and Transformation in Power Query (20 Marks)

The dataset was loaded into Power BI and directly transformed.



Type	Days_for_shipping_real	Days_for_shipment_scheduled	Benefit_per_order	Sales_per_customer	Delivery_Status	Late_deliver
DEBIT	3	4	91.25	314.6400146	Advance shipping	
TRANSFER	5	4	-249.0899963	311.3599854	Late delivery	
CASH	4	4	-247.7799988	309.7200012	Shipping on time	
DEBIT	3	4	22.86000061	304.8099976	Advance shipping	
PAYMENT	2	4	134.2100067	298.25	Advance shipping	
TRANSFER	6	4	18.57999992	294.980011	Shipping canceled	
DEBIT	2	1	95.18000031	288.4200134	Late delivery	
TRANSFER	2	1	68.43000031	285.1400146	Late delivery	
CASH	3	2	133.7200012	278.5899963	Late delivery	
CASH	2	1	132.1499939	275.3099976	Late delivery	
TRANSFER	6	2	130.5800018	272.0299988	Shipping canceled	
TRANSFER	5	2	45.68999863	268.7600098	Late delivery	
TRANSFER	4	2	21.76000023	262.2000122	Late delivery	
DEBIT	2	1	24.57999992	245.8099976	Late delivery	
TRANSFER	2	1	16.38999939	327.75	Late delivery	
DEBIT	2	1	-259.5799866	324.4700012	Late delivery	
PAYMENT	5	2	-246.3600006	321.2000122	Late delivery	
CASH	2	1	23.84000015	317.9200134	Late delivery	
DEBIT	2	1	102.2600021	314.6400146	Late delivery	
PAYMENT	0	0	87.18000031	311.3599854	Shipping on time	

The columns:

- Category_Id
- Customer_Email
- Customer_Fname
- Customer_Id
- Customer_Lname
- Customer_Password
- Customer_State
- Customer_Street
- Customer_Zipcode
- Department_Id
- Latitude
- Longitude
- Order_Customer_Id
- Order_Id
- Order_Item_Cardprod Id
- Order_Item_Id
- Order_State
- Order_Zipcode
- Product_Card_Id
- Product_Category_Id
- Product_Description
- Product_Image
- Product_Status
- Late_delivery_risk

Are not needed for the analysis hence are removed.

The columns Benefit_per_order and Order_Profit_Per_Order are computing the same values thus the former is dropped. The Order_City and Order_Country columns have values of FALSE which indicate that these are null. Hence, these values in these columns are dropped by filtering out.

Duplicates rows and rows with errors are dropped across the entire dataset. Power Query correctly identified every column thus no further intervention was needed. Furthermore, columns were renamed using capitalized first letters and spaces (Column Name) consistently throughout the entire dataset.

The Payment Type and Delivery Status columns' values were transformed so that only the first letter of each word was capitalized.The Customer Country used the abbreviated Spanish convention for the United States of America (EE. UU.) thus a remap and replace was performed. The Market column used the abbreviations USCA and LATAM which were mapped to United States and Canada and Latin America respectively. The Order Status column used all caps snake and camel case values. These were standardized to the first letter of each word capitalized and spaces between words. The Shipping Date and Order Date columns were remapped to use the US Date/Time locale of Month/Day/Year Time.

Finally dimension tables were created:

- DimPaymentType
- DimDeliveryStatus
- DimCategory
- DimCustomerCity
- DimCustomerCountry
- DimCustomerSegment
- DimDepartment
- DimMarket
- DimOrderCity
- DimOrderCountry
- DimOrderDate
- DimOrderRegion
- DimOrderStatus
- DimProduct
- DimShippingDate
- DimShippingMode

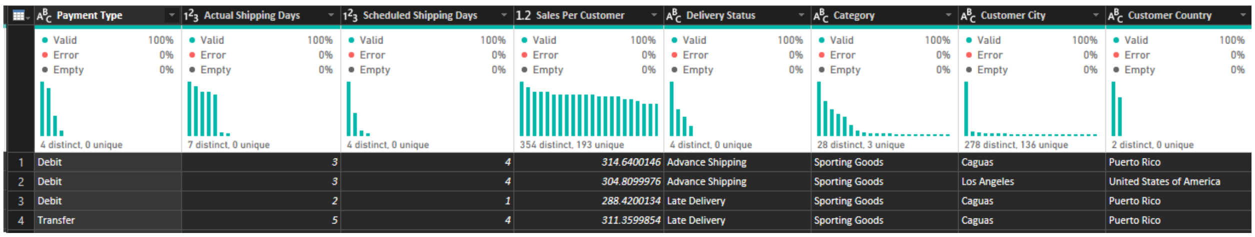
Within the DimOrderDate and DimShippingMode tables the month number, month name, day name, day number of the week, quarter, and year were extracted. The month number and day number of the week will be used to order their corresponding text pairs to ensure a deterministic appearance of values in the visuals.

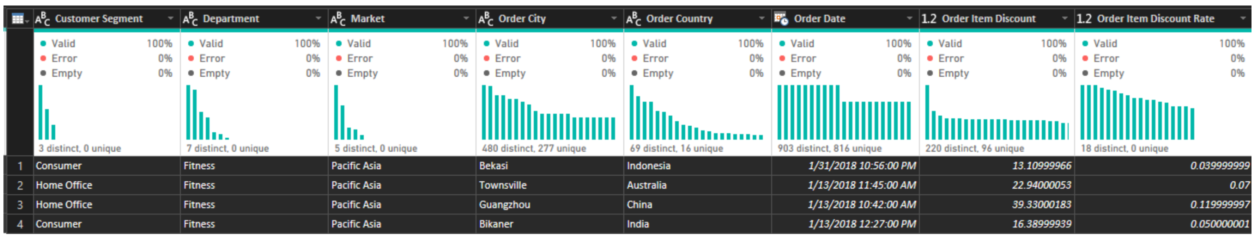
In summary, the following transformations are performed:

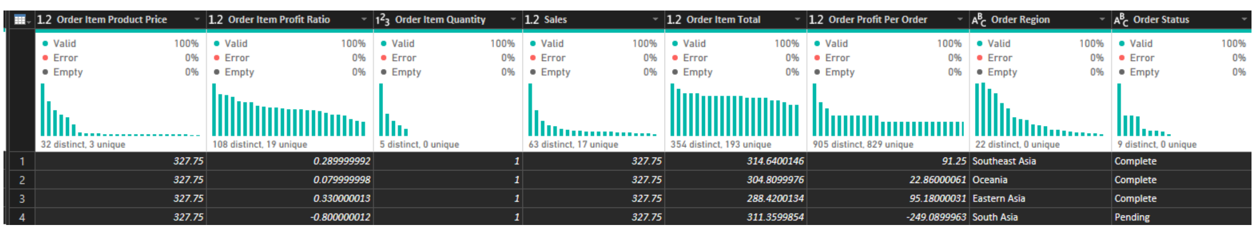
1. Remove unneccessary columns: To reduce the memory footprint and enhance query performance. High-cardinality or sensitive fields do not contribute to aggregate supply chain analysis and clutter the data model.
2. Remove duplicates: To ensure statistical integrity. Duplicate rows would lead to double counting of sales and profits, artificially inflating the performance metrics and leading to inaccurate business insights.
3. Remove errors: To prevent calculation failures. Power Query's error removal ensures that DAX measures and Power BI visuals don't break when encountering incompatible data types or corrupted cell values.
4. Remove null values: To maintain data quality in geographic reporting. By filtering out "FALSE" values in Order_City and Order_Country, it ensures that map visuals and regional breakdowns only reflect verified locations rather than "null" placeholders.
5. Rename Columns: To improve end-user readability and usability. Moving from technical snake_case to natural language makes the report intuitive for non-technical stakeholders to build their own visuals.
6. Value standardization: To ensure data consistency across categories. Standardizing inconsistent casing prevents the engine from treating "PENDING" and "Pending" as two different statuses, which would split the data incorrectly in charts.
7. Date/Time typecasting: To enable time-intelligence features. Converting dates to a standard US Locale allows Power BI to recognize the fields as chronological data rather than text, enabling year-over-year comparisons and trend lines.
8. Merging tables: To enrich data with external context. Using MapCustomerCountry and MapMarket transforms cryptic abbreviations into full, recognizable names, providing better context for regional performance analysis.

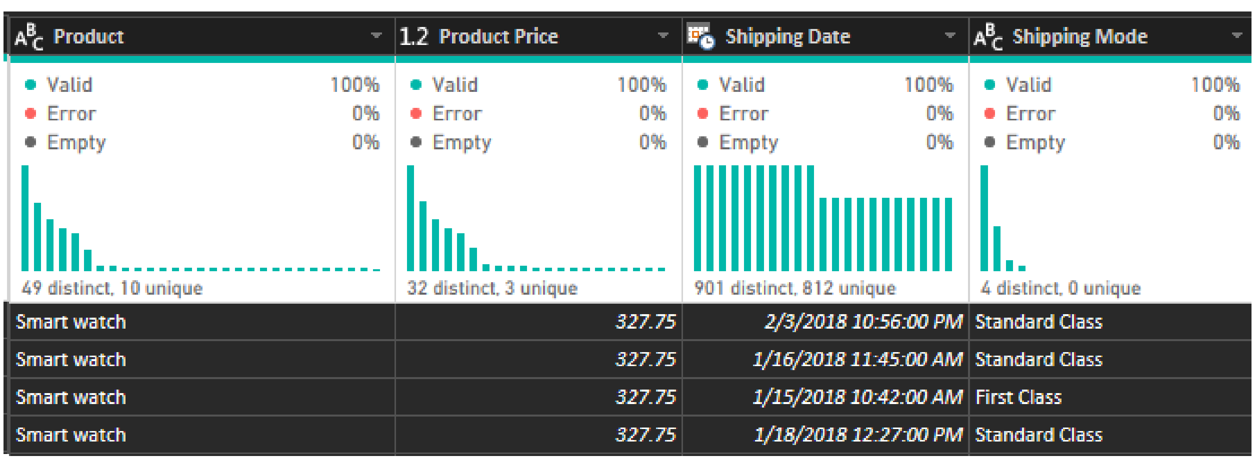
9. Extracting date parts: To support granular time slicing and logical sorting. Extracting attributes like Quarter, Month Name, and Day of Week allows for seasonal analysis. Crucially, keeping Month Number ensures that months appear in chronological order rather than alphabetical order in reports.

The Power Query screenshots before cleaning are in Part A. After cleaning the main Fact table became:

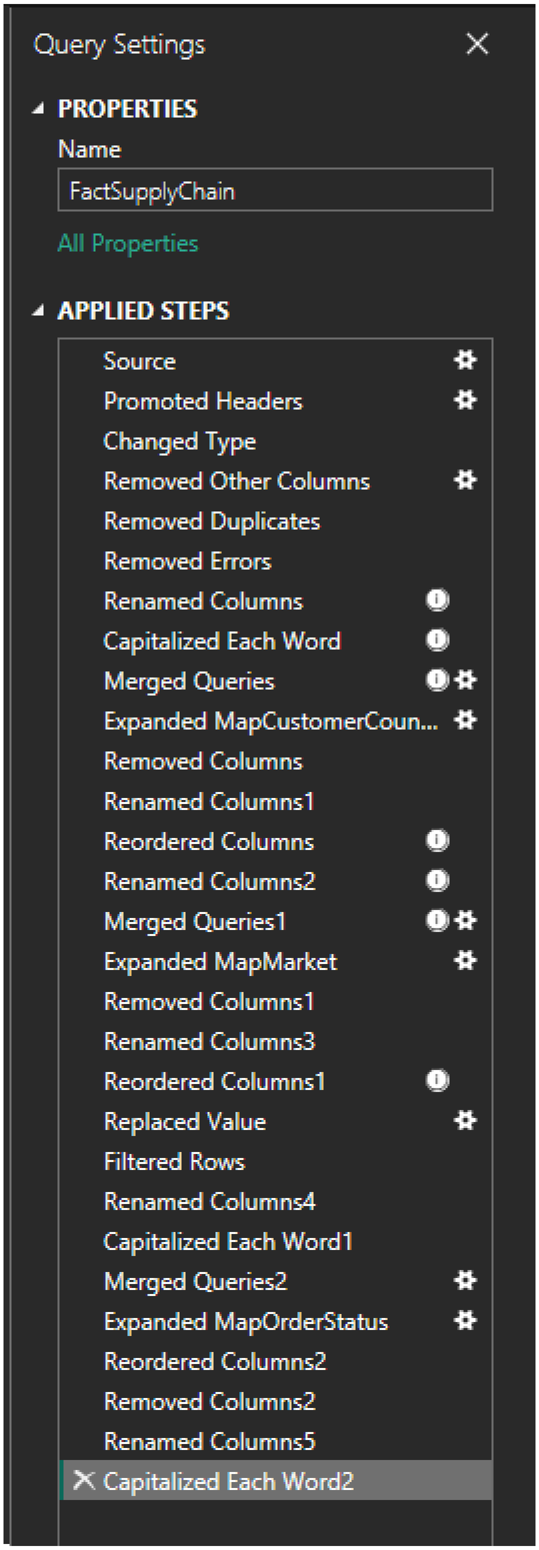




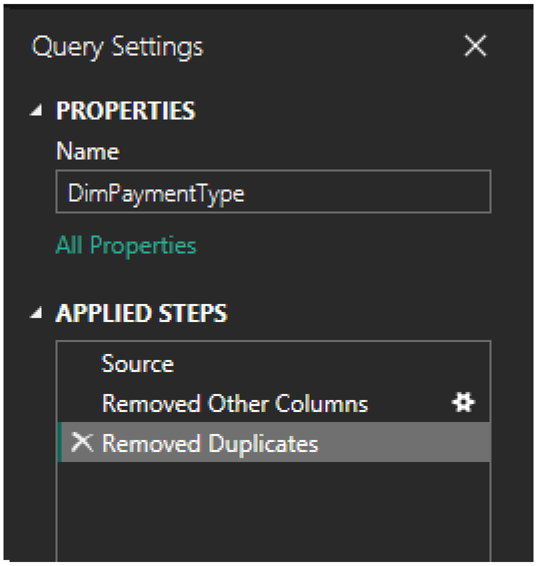




The Power Query applied the following transformations:

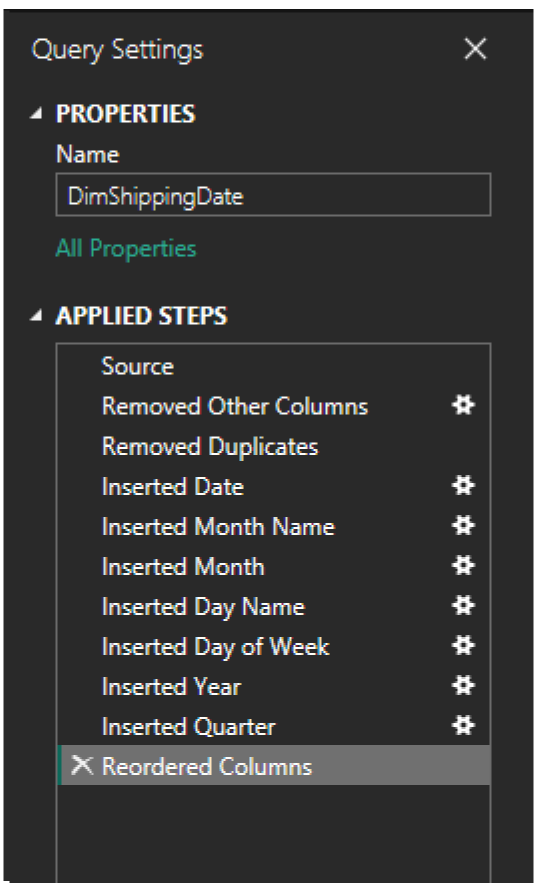


For every dimension table except DimOrderDate and DimShippingDate the Power Query pane followed this pattern:



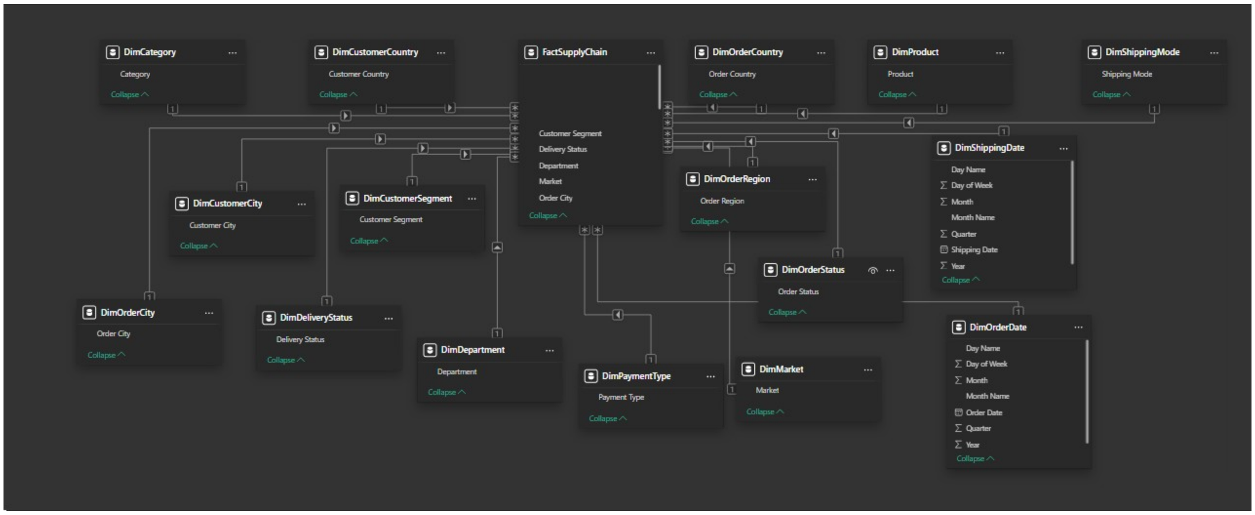
Other columns were removed and only unique values were retained for the column that the dimension table scoped.

For DimOrderDate and DimShippingDate time scoped transformations were applied:



Part C: Data Modeling (15 Marks)

The data model relationships are:



The architecture of the data model is built upon the Star Schema principle, which serves as the industry standard for high-performance analytical databases. At the heart of this structure lies the FactSupplyChain table, which is the primary source of truth for all quantitative measurements. This table is justified as a fact table because it houses the numerical metrics that are analyzed across different business perspectives. It acts as the central pivot point where all transactional events are recorded, connecting various business processes into a single, cohesive dataset.

Surrounding this central hub are the Dimension Tables, such as DimProduct, and DimDepartment. These tables are essential because they provide the descriptive context for every record in the fact table. They are justified as dimensions because they contain relatively static, qualitative attributes used to slice, dice, and filter the metrics.

The dimension tables have the following roles:

- DimPaymentType: Stores the different methods buyers used to pay for the goods they wanted shipped to them.
- DimDeliveryStatus: Stores the status of the requested orders. Some arrived early, others late, others on time, and some were canceled.
- DimCategory: Stores the category to which ordered products belong.

- DimCustomerCity: Stores the city from which a customer ordered a product which is simulteanously the dispatch destination of the ordered product.
- DimCustomerCountry: Stores the country from which a customer ordered a product which is simulteanously the dispatch destination of the ordered product.
- DimCustomerSegment: Stores the segment of the entity ordering some goods. These are individuals or large corporate organizations.
- DimDepartment: Stores the department to which the product ordered belongs to.
- DimMarket: Stores the geographical market from which the order was sourced.
- DimOrderCity: Stores the city from which goods were dispatched.
- DimOrderCountry: Stores the country from which goods were dispatched.
- DimOrderDate: Stores the date and time at which goods were requested for shipment.
- DimOrderRegion: Stores the region from which goods were dispatched.
- DimOrderStatus: Stores the status of the requested order. Some were successful while others were suspended for various reasons.
- DimProduct: Stores the name of product ordered.
- DimShippingDate: Stores the date and time at which goods arrived at the shipping destination.
- DimShippingMode: Stores the mode used to deliver an order. Some orders were rapid with same day delivery while others had a low second class priority.

To ensure the model remains performant and logical, One-to-Many (1: *) relationships have been established between the dimensions and the fact table. This cardinality is correct because a single attribute, like a specific "Market" in DimMarket, can be associated with thousands of different sales transactions in the fact table, but each transaction belongs to only one specific market. Furthermore, the Cross-Filter Direction is set to "Single," meaning the filter flow moves exclusively from the dimension tables to the fact table. This prevents the ambiguous relationship errors that often plague complex models and ensures that calculations like Year-to-Date totals are processed with maximum efficiency by the Power BI engine.

Temporal analysis is handled through a specialized implementation of Date Tables, specifically DimOrderDate and DimShippingDate. By relating these two distinct tables to the Order Date and Shipping Date foreign keys in the fact table, the model enables the analysis of performance across different timelines simultaneously. This allows for complex reporting, such as comparing the monthly sales growth against the monthly shipping volume without one calculation interfering with the other.

Ultimately, this specific model structure is designed to support efficient reporting by optimizing how data is stored and retrieved. By normalizing descriptive data into dimensions, the model reduces data redundancy, which keeps the overall file size small and the memory usage low.

Part D: DAX Measures and Calculated Columns (20 Marks)

The calculated columns are Order Value Tier and Profitability Segment. The Order Value Tier calculated column allows marketing teams to segment and target high-spending customers for loyalty programs. This enables them to discover the customer segment that spends the most and focus efforts of relationship management for greater retention. In this case, high values are those that are above 650 (given a maximum of 1500 and average of 255.68).

```
Order Value Tier =
IF(
    'FactSupplyChain'[Sales] > 650,
    "High Value",
    "Standard Value"
)
```

The Profitability Segment calculated column is invaluable for root cause analysis. By selecting "Loss Making," an analyst can immediately see which specific products, markets, or shipping modes are consistently losing money. This allows the business to stop selling certain products in specific regions or renegotiate shipping contracts that are eating into the margins.

```
Profitability Segment =
SWITCH(
    TRUE(),
    'FactSupplyChain'[Order Item Profit Ratio] < 0, "Loss Making",
    'FactSupplyChain'[Order Item Profit Ratio] <= 0.20, "Low Margin",
    'FactSupplyChain'[Order Item Profit Ratio] <= 0.50, "Medium Margin",
    "High Margin"
)
```

The Total Sales DAX measure computes the total amount made by the shipping company. This enables the company to get their overall sales and when coupled with a slicers show the total sales for a dimension or group of dimensions. This is the primary KPI for tracking the overall scale of the business.

```
Total Sales = SUM('FactSupplyChain'[Sales])
```

The Average Order Value DAX measure computes the average of all sales and when coupled with a slicers show the average sales for a dimension or group of dimensions. Helps determine if customers are ordering more expensive items over time or if bundles are increasing the basket size.

```
Average Order Value = AVERAGE('FactSupplyChain'[Sales])
```

The Unique Customers DAX measure counts the unique occurrences of customers (proxied by city). Monitors market reach and customer acquisition success. It evaluates the number of jurisdictions that the company has penetrated into.

```
Unique Customers = DISTINCTCOUNT('FactSupplyChain'[Customer City])
```

The Total Profit DAX measure aggregates the net profit after all costs and discounts. It is vital for assessing the financial health of the supply chain, as high sales don't always mean high profit.

```
Total Profit = SUM('FactSupplyChain'[Order Profit Per Order])
```

The Profit Margin % expresses profit as a percentage of sales using the DIVIDE function to safely handle zero-denominator errors. It identifies which products or regions are the most efficient at generating bottom-line value.

```
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
```

The Total Sales YTD DAX measure accumulates sales from the beginning of the current calendar year up to the latest date. It is critical for tracking performance against annual targets and identifying if the company is on track for year-end goals.

```
Total Sales YTD = TOTALYTD([Total Sales], 'DimOrderDate'[Date])
```

The Previous Year Sales DAX measure looks back at the same timeframe in the previous year and computes the total sales. It provides the baseline for growth analysis, allowing the evaluation of whether the company is performing better or worse than the previous cycle.

```
Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('DimOrderDate'[Date]))
```

The Sales Growth % DAX measure uses a variable to calculate the percentage change between the current period and the previous year. This is the ultimate trend indicator. Positive growth signals expansion, while negative growth flags a need for strategic intervention.

```
Sales Growth % =
VAR SalesLY = [Sales LY]
RETURN
DIVIDE([Total Sales] - SalesLY, SalesLY, 0)
```

Part E: Dashboard Design and Visualization (20 Marks)

Slicers

The slicers will be the dimensions:

- Category as button slicer
- Customer Segment as a button slicer
- Delivery Status as a button slicer
- Department as a button slicer
- Market as a button slicer
- Order Date as a slider slicer
- Order Region as button slicer
- Order Status as a button slicer
- Shipping Date as a slider slicer
- Shipping Mode as a button slicer
- Order Value Tier as a button slicer
- Profitability Segment as a button slicer

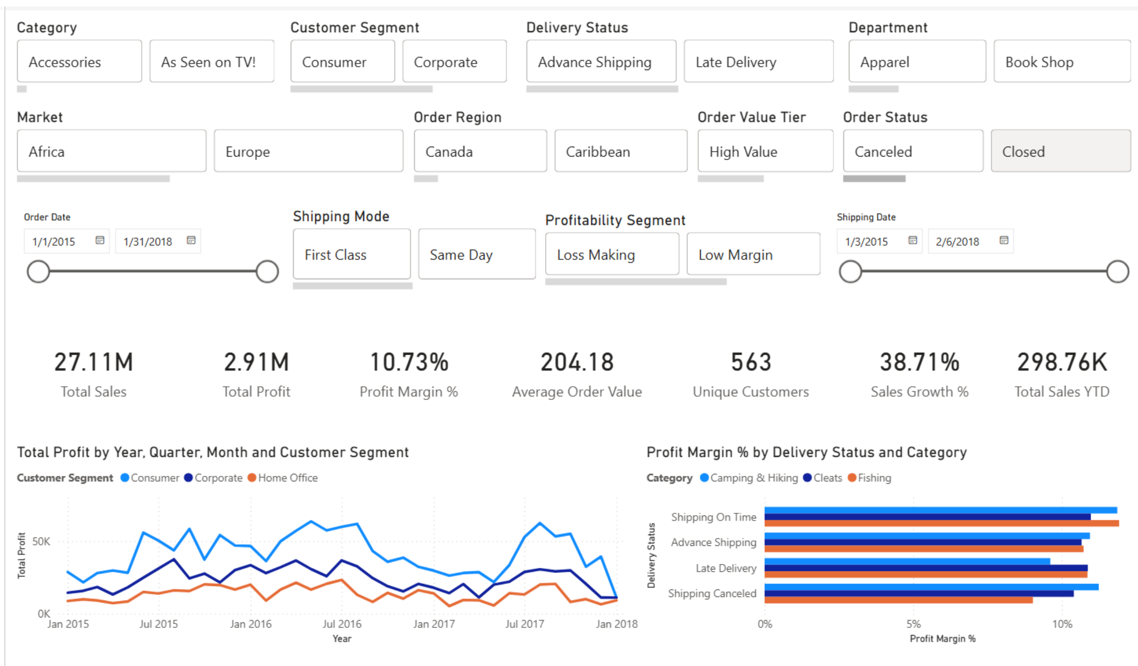
The Customer Country and Order Country dimensions are not included as slicers since a map visual will be used that is interactive thus effectively acting as a slicer.

Design Choices

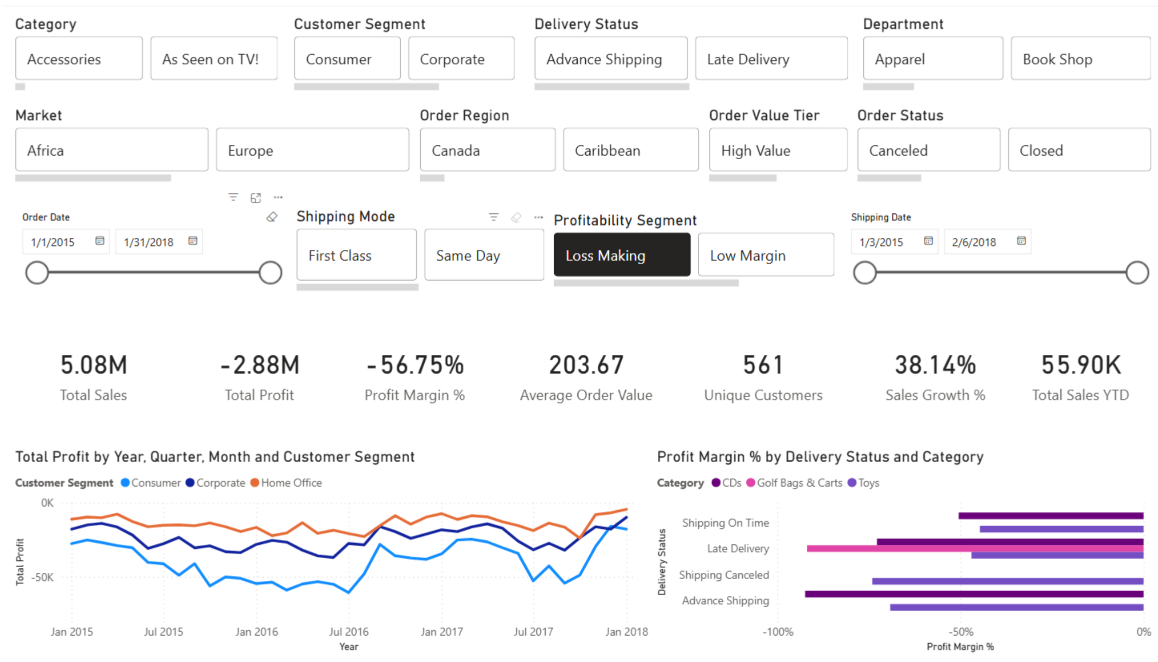
On the first page the slicers will be at the very top for visuals to be filtered by. These slicers are synchronised with all the visuals on all pages thus any filtering or drill down logic perpetuates across all pages.

Executive Summary

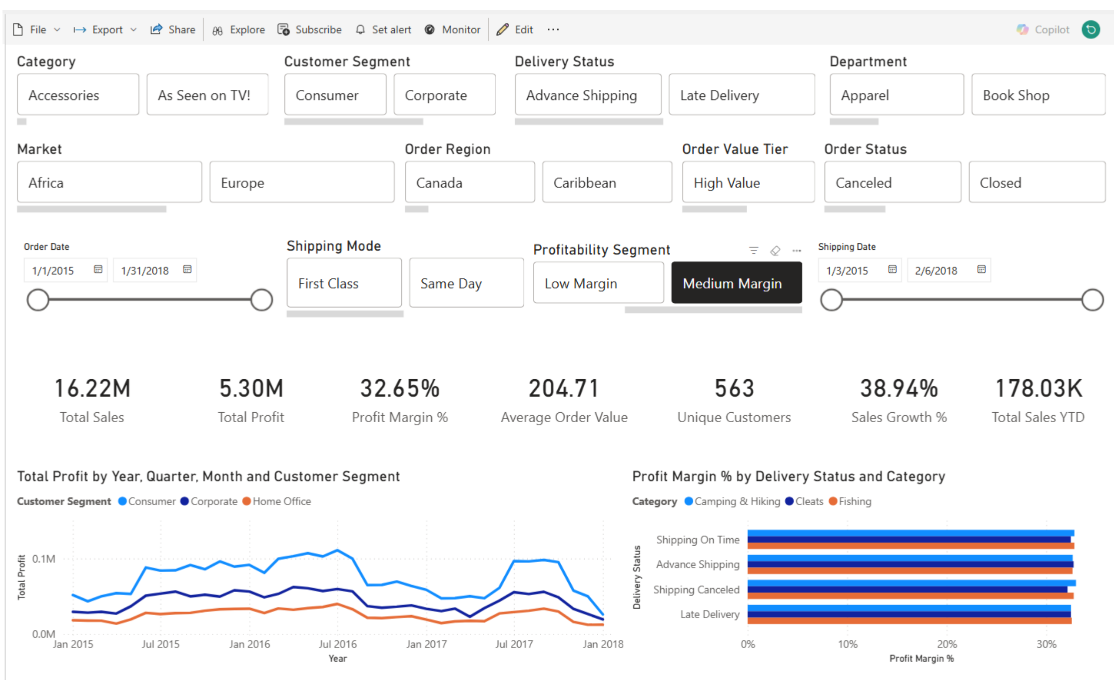
The executive summary provides KPI cards, a trend chart, category comparison and slicers.



A user interacts with the dashboard using the slicers provided across the various dimensions. For example, a user may want to examine the profit (loss) margins of the company makes when they are losing money. In this scenario the company makes a loss in 464 customer countries experiences a loss margin of 55.23% but also has a sales growth of 35.87% indicating the company has an upward trend in mitigating their losses viz a viz the previous year. This shows that the company has put in place strategies that are working to minimise their losses.

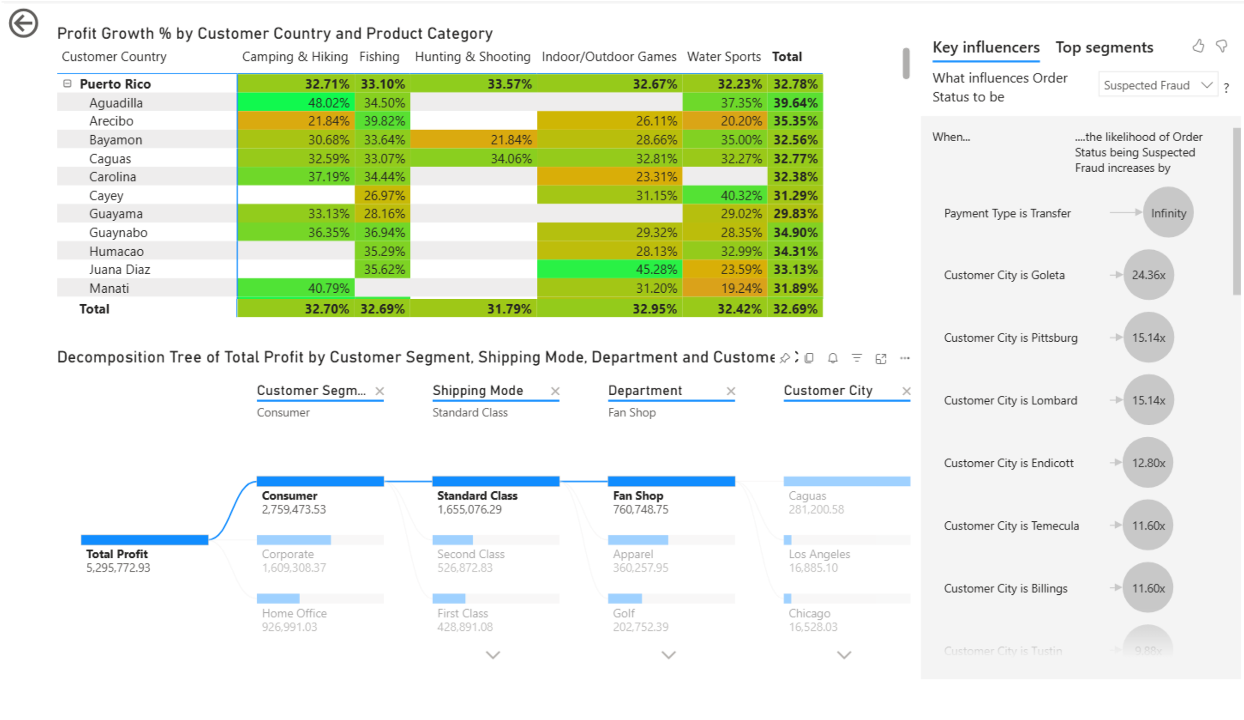


On the other hand, when the company has medium profit margins it experiences this profit across 563 jurisdictions with an average profit margin of 32.65% which is quite modest and a higher sales growth percentage of 38.94%. This clearly shows that this company is putting in place strategies that are resulting in a net positive growth.



Detailed Analysis

The detailed analysis page seeks to uncover the patterns in the order status, total profit, and profit growth decomposed across the dimensions of customer segment, shipping mode, department, and customer city.

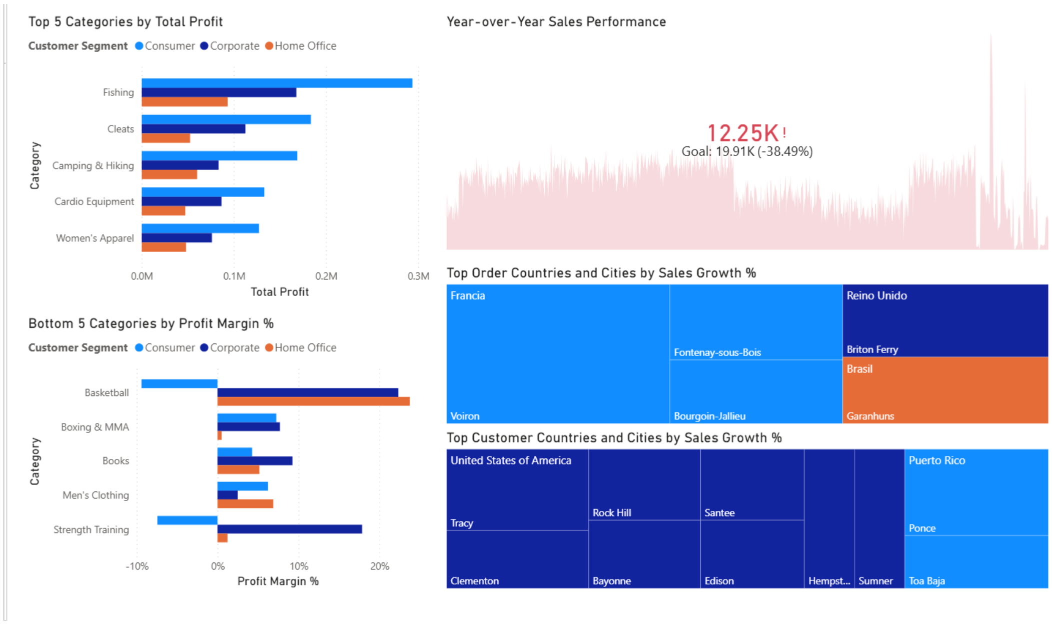


In the above scenario the key influencers of a suspected fraud are being evaluated within the customer cities within the consumer customer segement, standard class shipping mode, and fan shop department. In this scenario, orders made from Goleta, Pittsburg and Lombard are likely to be fraudulent transactions.

Insights and Performance Monitoring

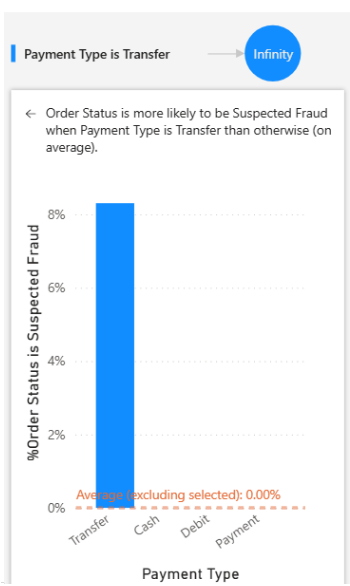
Within the insights and performance monitoring the top and bottom 5 categories are displayed. The top 5 categories are determined by the total profit while the bottom 5 categories are determined by the profit margin percentage.

In this case it is evident that losses are made when consumers purchase products from the Basketball and Strength Training categories. Otherwise, there is a general positive profit margin. The Year-Over-Year sales performance compares the current viz a viz the previous year to determine if there has been a positive increase in the sales from the previous year. In the scenario below there is a decrease in the sales made.

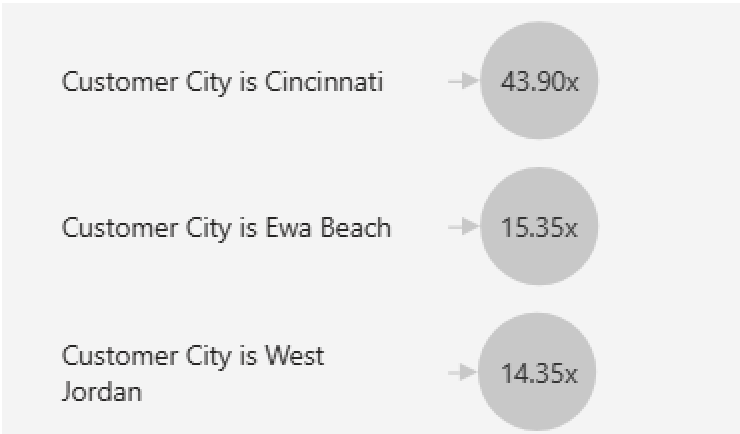


Part F: Analytical Insights and Business Recommendations (10 Marks)

First of all, across the board when suspected fraud transpires the only payment method done is a bank transfer.

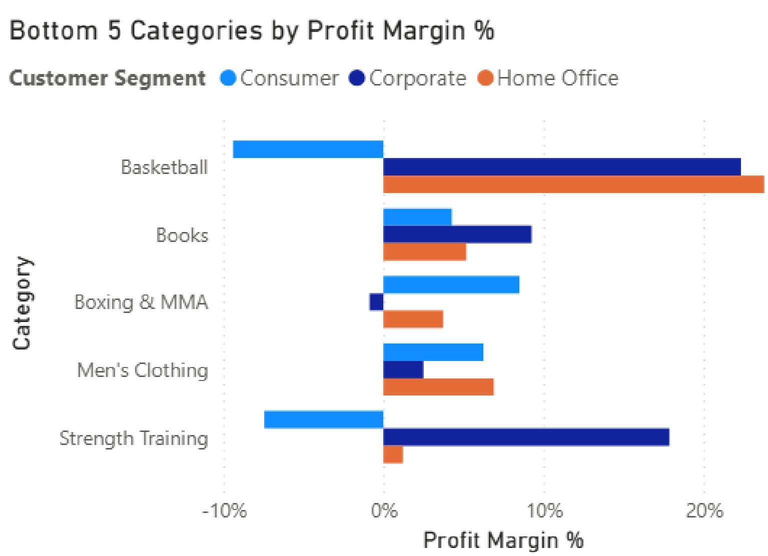


Secondly, the cities that are on the radar for having fraudulent orders are Cincinnati, Ewa Beach and West Jordan respectively.

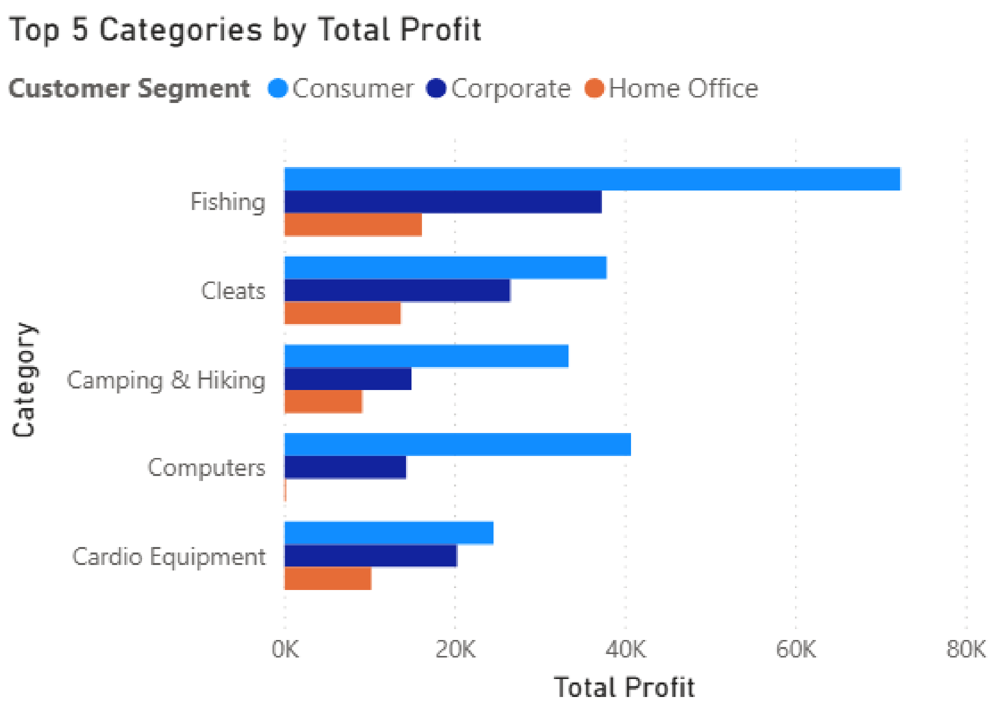


Therefore, it is imperative that extra scrutiny is placed on all orders that are attempting to select transfer as the payment method. On top of this, if these orders originate from the aforementioned three cities the company should exercise even greater care in engaging with these orders to minimise the probability of losing money on shipping a fraudulent order.

Thirdly, in the last year of business (31st January 2017 to 31st January 2018) the business experienced losses on the Basketball and Strength Training categories for the Consumer customer segment. Moreover, in the Corporate customer segment for Boxing & MMA the company lost some money.



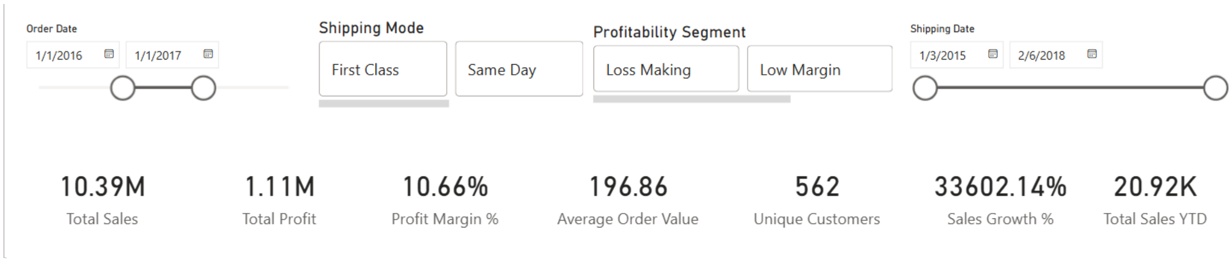
Fourthly, the company should focus on selling products within the Fishing, Cleats and Camping & Hiking categories especially. These categories have modest profits across the board across all customer segments.

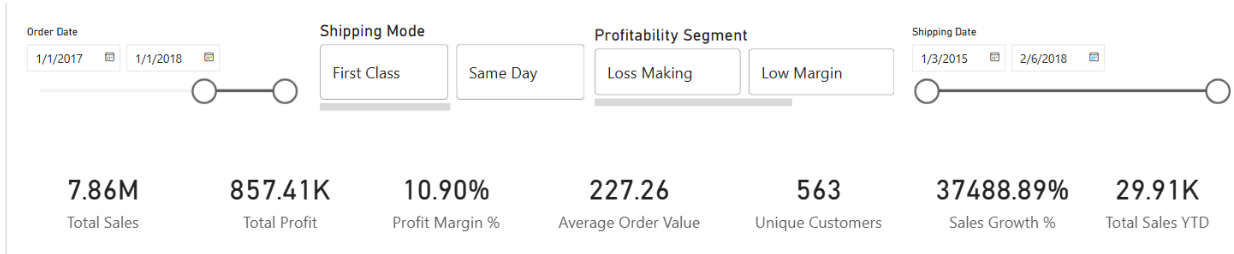


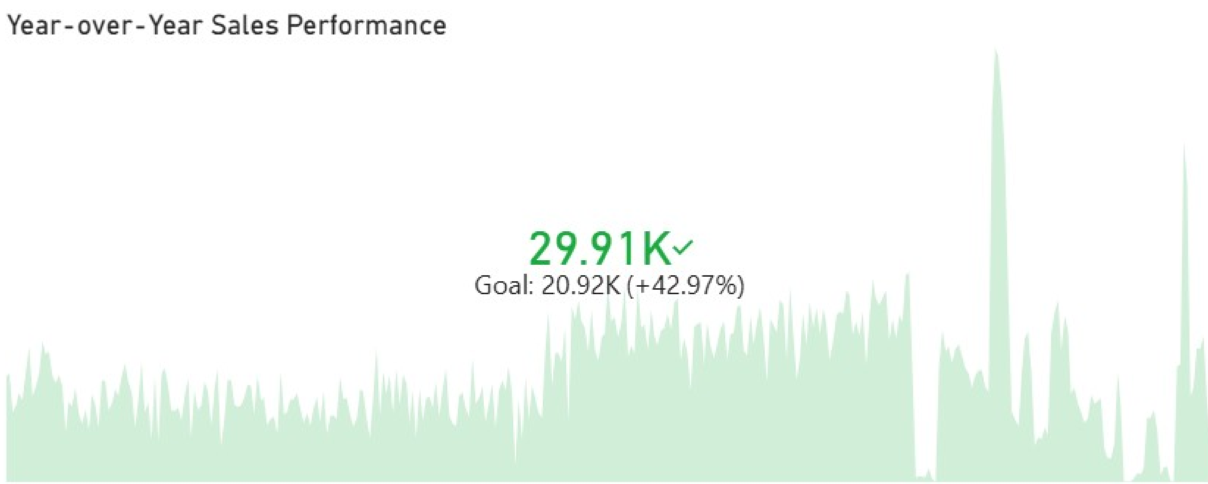
Thus, it would be prudent for the company to pivot away entirely from selling products in the categories it is losing money or sell only the products that are profitable within the categories to ensure that it maximises its profits. Moreover, they should redouble their efforts in marketing products in the Fishing, Cleats, and Camping & Hiking categories. These are their bread and butter.

Fifthly, when comparing sales growth year over year (2016 - 2017 and 2017 - 2018 with 2015 - 2016 as a buffer year) it is evident that although the company does not make as many sales in the later year (2017 -2018) they greatly increase their sales growth, their average order value and profit margins. These standardized values show that the company implemented policies that worked positively in improving their margins. They likely focused on products that guaranteed larger profit margins and ditched those that did not perform so well.

A cross comparison with the Year-over-Year Sales Performance visual shows that in the calendar year 2017 - 2018 the sales performance dominated the previous year's performance (2016 - 2017) by an increase of 42.97%.

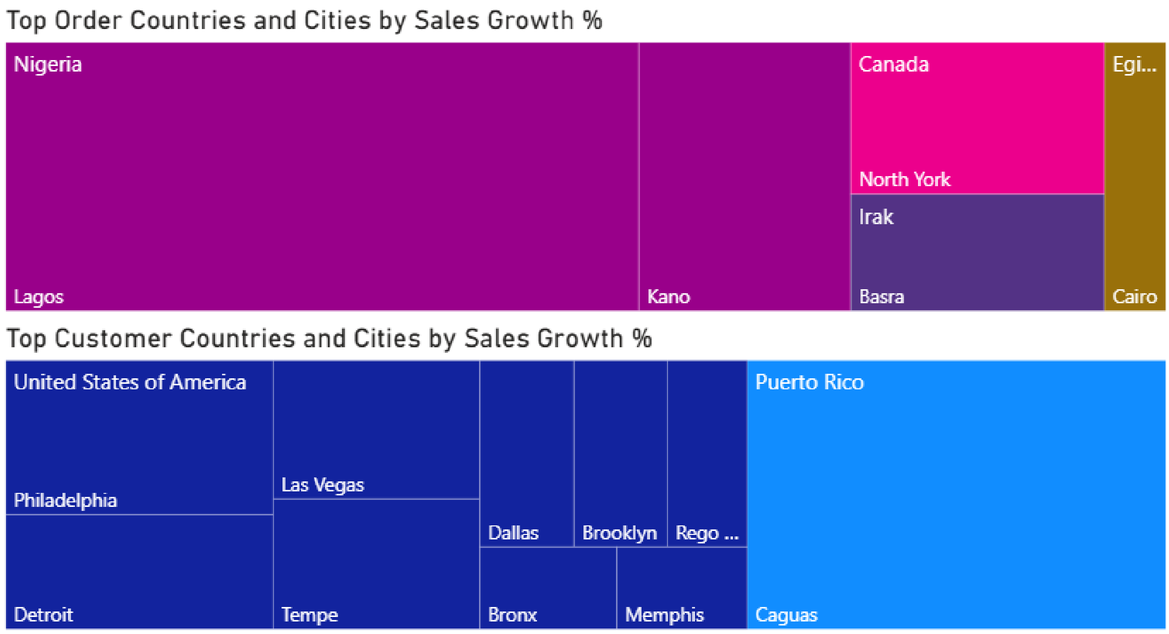


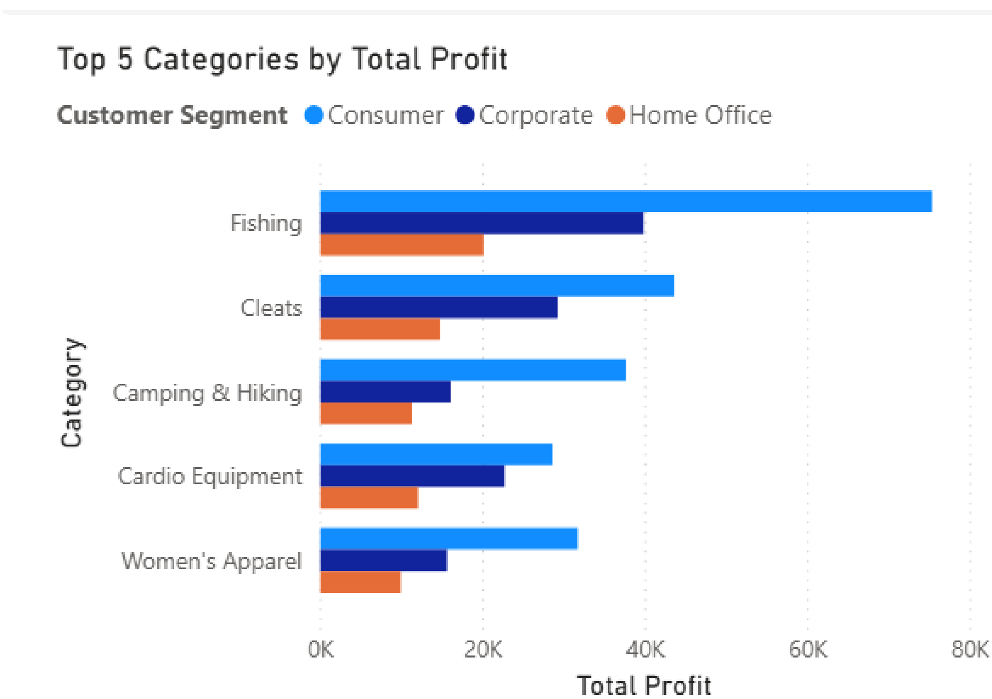


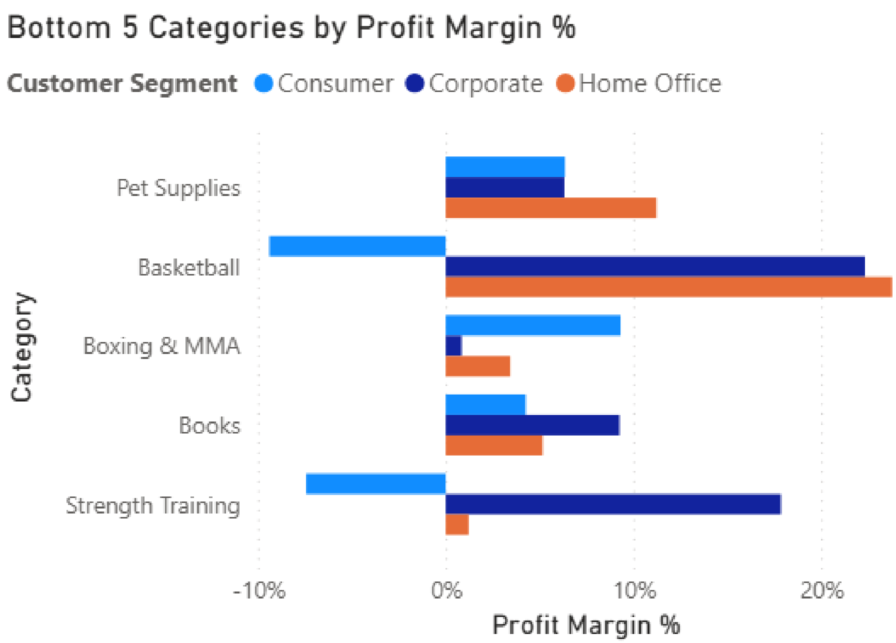


As a result, I recommend that the company maintains its current trajectory and incorporates the insights that in this recent year (2017 - 2018) the products causing them to lose money are Basketball and Strength Training products sold to Consumers. Moreover, they should focus on selling Fishing, Cleats, and Camping & Hiking products across their different customer segments but especially to Consumers. Furthermore, the countries from which products with the largest sales growth were sourced from are Nigeria and Canada. There is likely a growing export market in these countries and the logistics company should position itself to exploit this opportunity.

Similarly, in the cities of Philadelphia, Detroit, and Las Vegas in the United States of America there is a budding market for the products the logistics company ships. In Puerto Rico Caguas is the locale that dominates the growth in sales to customers. Special focus should be given to these cities.







Part G: GitHub Repository and Documentation (5 Marks)

In my analysis of this global supply chain, I've moved beyond simple surface metrics to identify the real friction points where logistical inefficiency meets financial risk. It's clear that a one-size-fits-all approach is no longer sustainable for the company. By digging into the intersections of payment behavior, regional growth, and product profitability, I have mapped out a strategy that doesn't just protect company margins but actively aggressively pursues the most efficient paths to scale.

One of my most critical findings was the absolute correlation between payment methods and suspected fraud. Every single instance of flagged fraudulent activity originated from Bank Transfers. This is a massive red flag. Furthermore, the data pinpointed a high-risk geographic cluster: Cincinnati, Ewa Beach, and West Jordan.

The company needs to adopt a proactive stance. Any order utilizing a bank transfer from these specific locales should trigger an immediate manual audit. By increasing scrutiny here, the company isn't just stopping bad orders; it is preserving the logistical resources that would have been wasted on shipping goods that never result in a legitimate sale.

Through my profit-margin heatmapping, I identified a significant disparity in how our product categories perform across different customer segments. While our overall sales numbers might look healthy, there is dead weight in the Basketball and Strength Training categories when sold to general consumers. These segments are currently loss makers that dilute the hard earned gains.

Conversely, I’ve identified our Bread and Butter segments: Fishing, Cleats, and Camping & Hiking. These categories show modest, resilient profits regardless of the customer segment or shipping mode.

The company should pivot its marketing spend away from low performing sports categories and redouble its efforts in Fishing and Cleats. The company need not sell everything to everyone; but rather to sell the right things to the right people to maximize the bottom line.

My Year-over-Year (YoY) analysis revealed a fascinating shift in the 2017-2018 calendar year. Even though the raw sales volume didn't always hit historic peaks, the Average Order Value and Sales Growth % improved. This indicates that the company’s recent policy shifts, likely focusing on higher-quality transactions over sheer volume, are working.

The data shows a burgeoning export market in Nigeria and Canada, and high density domestic growth in Philadelphia, Detroit, and Las Vegas. These are the company’s high-velocity markets. There should be a prioritization on these lanes for infrastructure investment. Specifically, in Puerto Rico, Caguas stands out as a dominant growth hub. Should the company position its logistics centers closer to these high growth cities, the company can decrease the Actual Shipping Days while maintaining the high profit margins seen in the latest cycle.

GitHub Link: https://github.com/mistiusiu/DSA3050_SS26_USIU_End_Semester_Exam